DANESHVAR AMROLLAHI

daneshvar@cs.stanford.edu cs.stanford.edu/~daneshva Google Scholar GitHub

EDUCATION

- **Stanford University** *2024/01 - Present*
  PhD in Computer Science *Advisor: Clark Barrett*
- **University of Tehran** *2018/09 - 2023/02*
  BSc in Computer Engineering (Software) *GPA: 18.02/20.00*

SELECTED PUBLICATIONS

- D. Amrollahi et al. **Faithful Autoformalization via Roundtrip Verification and Repair.** *Findings of the Association for Computational Linguistics: EMNLP 2026. (Workshop: AI4Math @ ICML 2026)*
- O. Wang et al. **Know Your Limits: On the Faithfulness of LLMs as Solvers and Autoformalizers in Legal Reasoning.** *Under review at ACL Rolling Review (ARR). (Workshops: AI4Math, AI4Law @ ICML 2026)*
- C. Sun et al. **VeriStruct: AI-assisted Automated Verification of Data-Structure Modules in Verus.** *International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2026)*
- D. Amrollahi et al. **Towards SMT Solver Stability via Input Normalization.** *Formal Methods in Computer-Aided Design (FMCAD 2025).*
- D. Amrollahi et al. **Solving Invariant Generation for Unsolvable Loops**. *International Static Analysis Symposium (SAS 2022).* **Awarded the Radhia Cousot Young Researcher Best Paper Award.**

INDUSTRY EXPERIENCE

- **PhD Software Engineer Intern at Uber Technologies, Inc.** *Seattle, United States*
  Programming Systems Team *2026/06 – 2026/09*
  Developed a **static analysis** for Go that detects **non-determinism** in Cadence workflows, a correctness bug class tests cannot reliably catch. Deployed in **shadow mode** on live pull requests and benchmarked against Uber's **LLM**-based code reviewer: comparable precision and recall, with deterministic results at zero inference cost.
- **Applied Science Intern at Amazon Web Services (AWS)** *Santa Clara, United States*
  Automated Reasoning Group *2025/06 – 2025/09*
  Built a tool from scratch for validating SMT-LIB arithmetic queries and models: integrated a Rust parser to construct Lean objects, developed a Rust–Lean FFI pipeline, encoded SMT terms and semantics in **Lean**, and formally proved correctness of optimized evaluation against reference semantics.

RESEARCH EXPERIENCE

- **PhD Student at Center for Automated Reasoning, Stanford University** *Stanford, United States*
  Under Prof. Clark Barrett *2024/01 – Present*
  1. Proposed a **roundtrip verification** and **repair** framework for **faithful autoformalization** of natural-language into first-order logic, without ground-truth annotations.
  2. Co-developed **Lean** formalizations of Python programs and curated tasks for **VeriBench**, an end-to-end formal verification benchmark for evaluating **LLMs** on verified code generation.
  3. Solo-led the design, implementation, and large-scale evaluation of a **normalization** algorithm in the **cvc5** industrial SMT solver. Proved the problem as hard as **graph isomorphism** and gave a practical approximation, evaluated across ~570K solver runs, significantly improving stability of **cvc5** and **Z3**.
- **Research Intern at Dependable Systems Lab, EPFL** *Lausanne, Switzerland*
  Under Prof. George Candea *2022/07 – 2022/08*
  Extended the KLEE **symbolic execution** engine to generate and track **quantified formulas** in **Z3**, enabling **loop summarization** to mitigate **path explosion** and improve scalability in analyzing **network software**. Involved substantial C++ **systems programming** and **SMT solver** integration.

- **Research Intern at Automated Program Reasoning Group, TU Wien** *Vienna, Austria*
  Under Prof. Laura Kovács and Prof. Ezio Bartocci *2021/07 - 2022/02 + 2023/05 - 2023/12*
  Developed a **program synthesis** system for recursive programs grounded in **theorem proving**, and a **static analysis** tool for synthesizing loop invariants using **symbolic reasoning** approaches.

HONORS AND AWARDS

- **Radhia Cousot Young Researcher Best Paper Award** *2022*
  29th Static Analysis Symposium (SAS 2022)
- **Stanford School of Engineering (SoE) Fellowship** *2024*
  Awarded a $12900/quarter stipend and full tuition coverage for one year
- **Ranked 8th** in **ACM-ICPC (International Collegiate Programming Contest)** *2020*
  West Asia Region, Tehran site
- **Summer@EPFL Fellowship** *2022*
  Ranked top 1.5% among 4000 applicants and awarded a 1600 CHF/month fellowship over summer

PROJECTS

- **cvc5** — github.com/cvc5/cvc5 *C++*
  Implemented an efficient and scalable normalization pre-processing pass in this industrial-strength SMT solver for achieving performance stability across semantics-preserving query mutations.
- **ARM Processor** — github.com/daneshvar-amrollahi/ARM *Verilog, ModelSim, Quartus*
  Implemented a comprehensive 5-stage pipelined ARM processor with data forwarding, hazard detection, cache memory, and SRAM controller.
- **Polar** — github.com/probing-lab/polar *Python, SymPy*
  Implemented a polynomial loop-invariant synthesizer for (probabilistic) unsolvable loops.

SKILLS

- **Programming Languages**:
  - Experienced in C, C++, Python.
  - Familiar with Rust, Go, Scala, Bash, Verilog, SystemVerilog.
- **Tools**: cvc5, Z3, Lean, KLEE, LaTeX.
- **Academic (Sub)review**: ACL Rolling Review (ARR) 2026, AI4Math @ ICML 2026, TACAS 2026, ISSAC 2024, Journal of Symbolic Computation.